Task 2: Information Sources Identification for Mfano Bora Africa Chatbot Knowledge Base

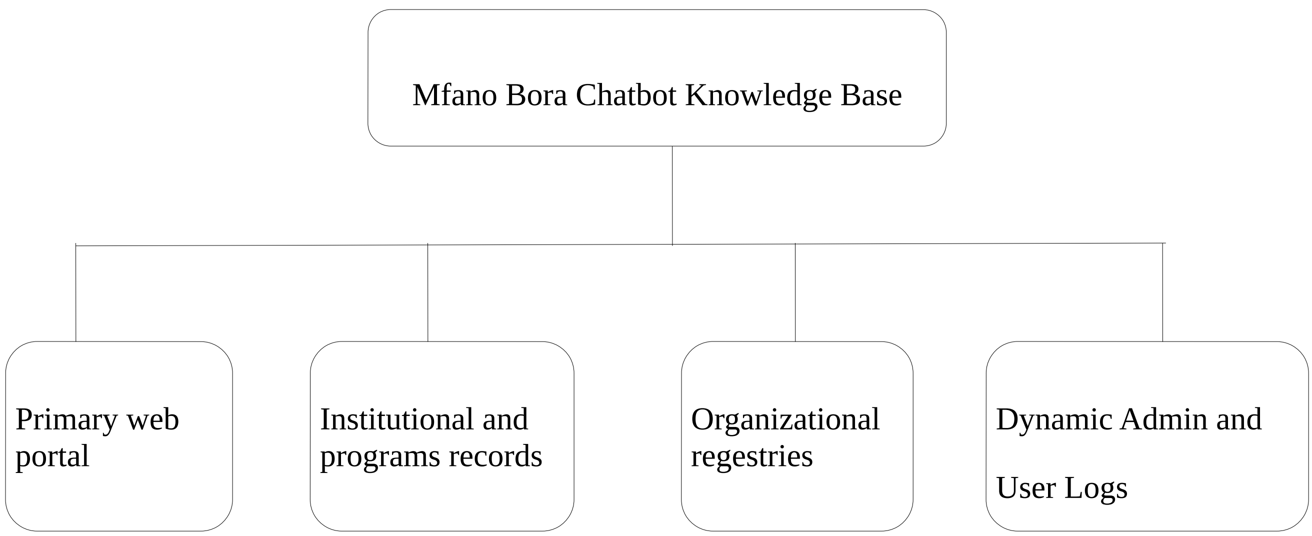
Introduction

In contemporary information management practices, creating an automatic dialogue system requires to have a trustworthy, verified, and structured set of facts. In the Retrieval-Augmented Generation RAG chatbot architecture, identifying the right information sources is important in overcoming any artificial intelligence "hallucinations," as well as achieving high precision in responses.

This paper outlines the process of identification and classification of all internal and external information sources that will be used in constructing the knowledge base for the Mfano Bora Africa Ltd Web-Integrated Chatbot. By incorporating verified facts about the corporation, attachment opportunities within academia, logistical consultancy services, and interaction logs, the knowledge base guarantees that any user inquiry whether from potential interns, corporate customers, or the community members will receive an accurate response or be referred to the appropriate web portal.

Information Sources Categorization for the Knowledge Base

For the creation of an single source of truth (SSOT), the information sources of the Mfano Bora Africa chatbot knowledge base are grouped into four important structural categories:



1. Primary Digital Sources (Official Web Portal)

The main digital sources constitute the basic foundation of the chatbot's conversation model. These web pages are will be regularly scraped, parsed, and converted into text chunks for vector indexing:

- Corporate and Services Portal: Comprehensive information on the company's core business operations such as logistics consultancy and integration of supply chain management frameworks.
- Careers and Attachment Portal: Application criteria and direct linkages to apply like https://www.mfanoboraafrica.com/apply-now/ for industrial attachment in ICT, logistics and administration.
- Institutional Governance and Policy Sources: Terms of service, privacy policies and website disclaimers providing the framework for user queries, boundaries and privacy notifications.

2. Institutional and Programs Records

Apart from static web content, Mfano Bora Africa runs a well-established industry programs and student development programs. The following documents from the institution provide contextual background information:

- Documentation on awards and programs: Historical documentation and records of the East Africa Transport, Logistics and Road Safety Awards that have been conducted for 24 years, along with regional programs undertaken by Road Safety Club Kenya.
- Testimonials and Syllabus from Industrial Attachments Students: Documented qualitative accounts from past industrial attachment students (including students from institutions such as the Cooperative University of Kenya and Technical University of Kenya) and syllabus from the institutions.

3. Oraganizations Operational and Contact Registries

Metadata related to the operations will enable the chatbot to answer instant queries on matters related to company operations:

- Physical Location List: Geographical location details (Mfano House, Ole Sein Rd, Nairobi, Kenya).
- Communication List: Means of direct communication such as e-mail addresses (info@mfanoboraafrica.com) and helplines.
- Standard Operational Hours: Operational timings (Monday to Friday: 7:00 AM – 5:00 PM; Saturday: 8:00 AM – 1:00 PM).

4. Dynamic Administrative and User Interaction Log Data

To achieve continuous evolution and flexibility in models, the chatbot uses live interaction data from its database as follows:

- Curated FAQ Databases: Pairs of questions and answers formulated by the Mfano Bora administrators for handling frequently asked customer questions.
- PostgreSQL Chat Logs (`chat_logs` and `admin_metrics`): Unanswered customer queries and failed intention matching captured in the database, which helps in identifying information gaps and updating knowledge base chunks (`knowledge_base` table).

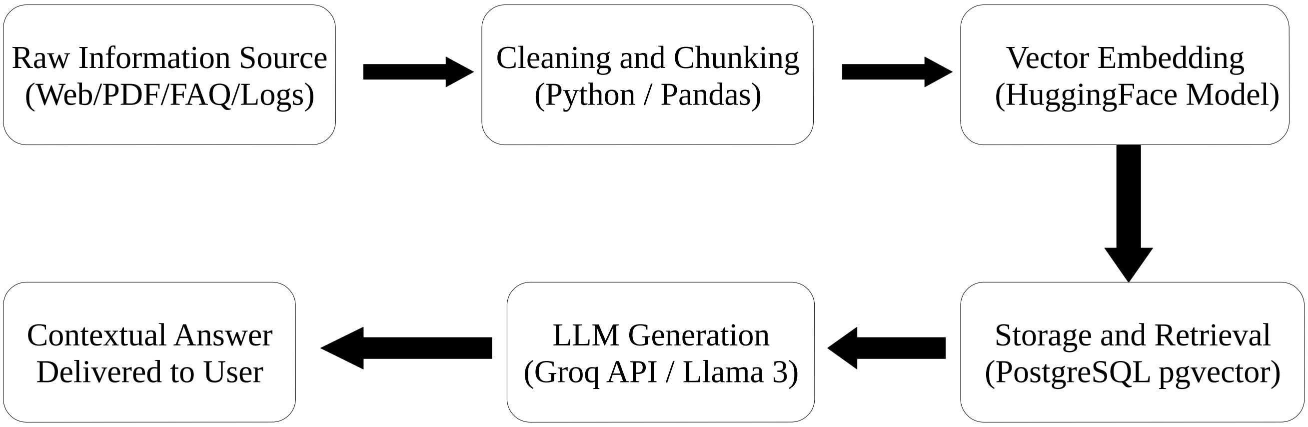
Information Sources Taxonomy and Knowledge Map Matrix

Below is a matrix that shows how each information source is mapped to the extracted knowledge content and user persona and update frequency.

Source Name	Source Type	Knowledge Base Content Extracted	Target User Group	Update Frequency
Mfano Bora Main Web Portal	Digital / Public	Core mission, supply chain management consultancy services, corporate history.	Corporate Clients & Logistics Prospects	Monthly
Attachment Portal (/apply-now/)	Digital / Application	Requirements for ICT, Logistics, and Admin attachments, computer packages, direct application link.	Students & Graduates	Quarterly
Awards & Road Safety Records	Internal / Event Doc	Event history, award categories, Road Safety Club participation guidelines.	Nominees, Industry Peers & Community Members	Bi-Annually
Location & Contact Directory	Metadata Registry	Physical address (Mfano House), office opening hours, official contact emails.	All Visiting Website Users	As Needed
Attachment Feedback Records	Qualitative Artifacts	Practical training modules, hardware/software troubleshooting experience, student expectations.	Prospective Interns & Attachment Applicants	Annually
Admin FAQ & Chat Audit Logs	Dynamic / Database	Frequently asked questions, fallbacks, unresolved queries (chat_logs).	System Administrators	Continuous / Real-Time

Data Acquisition and Ingestion Protocol

To integrate these identified sources into the chatbot's machine learning backend without incurring hallucination, an automated Data Acquisition Pipeline is implemented:



1. Extraction: Using web scraping libraries and admin ingestion scripts, text data is extracted from external sources like web portals and internal document repositories.
2. Chunks & Cleaning of Text: The raw text is cleaned by removing duplicate HTML tags and old data, and then is structured into chunks.
3. Vectorization: The chunks are converted to vector form using HuggingFace's Sentence Transformers.
4. Indexing with Vectors: The vectors are stored directly in PostgreSQL database using pgvector extension in knowledge_base table.
5. Retrieval & Parsing with LLM: If a user enters a query, then the backend runs a similarity search on `pgvector`, and passes only the relevant context chunks to Groq API (Llama-3 model).

References

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